UNIVERSITY OF OSLO

A Self-Adaptive Digital Twin Architecture for Dynamic Resource Management

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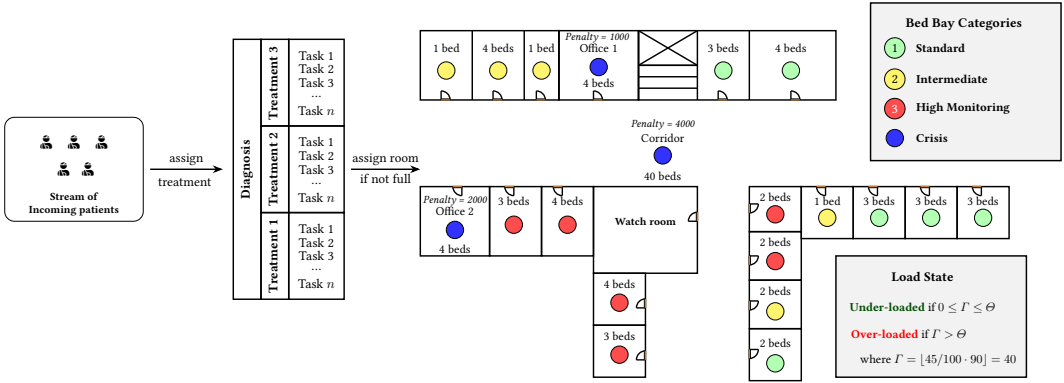
SEAMS 2026

April 13, 2026

Contents

Why do we need it?



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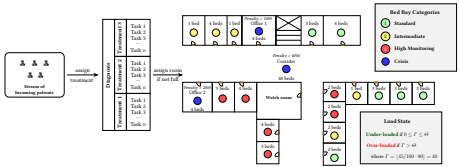

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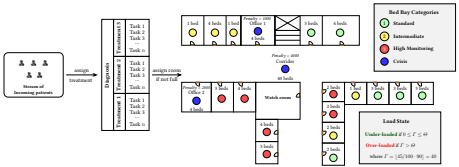

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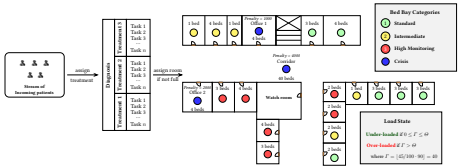

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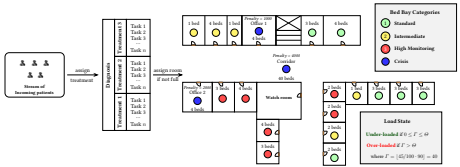

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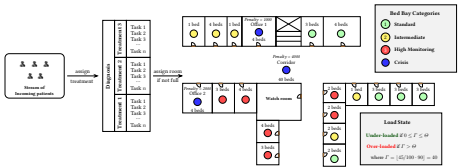

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- Static architectures struggle when the underlying structure changes
- We need runtime adaptation with trustworthy model alignment

Baseline: BedreFlyt and the gap

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- Human-in-the-loop actions can diverge from DT suggestions
- Main gap: the original ward layout is treated as fixed

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- Human-in-the-loop supervision over the adaptation loop

Main contributions

- **Self-adaptive DT architecture:** integrates semantic reflection and runtime orchestration for dynamic ward structure changes

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- **Penalty-based optimisation model:** supports adaptation decisions by balancing feasibility and adaptation cost
- **Formalised adaptation objective:** ensure patient allocation feasibility with the fewest rooms while minimising penalty
- **Human-supervised autonomy:** operators approve actions while the DT keeps structural and behavioural models aligned

Contents

What is a Digital Twin

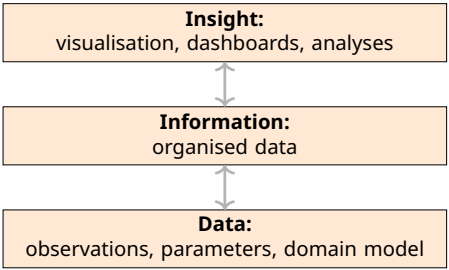
Digital twins are, as defined by NASEM

NASEM Definition of DT (2024)

A digital twin is a set of virtual information constructs that mimics the structure, context, and behaviour of a natural, engineered, or **social system** (or system-of-systems), is dynamically updated with data from its physical twin, has a predictive capability, and **informs decisions** that realise value. The bidirectional interaction between the virtual and the physical is central to the digital twin[a].

[a] National Academies of Sciences, Engineering, and Medicine (NASEM) (2024): Foundational Research Gaps and Future Directions for Digital Twins. The National Academies Press

Digital Twins components



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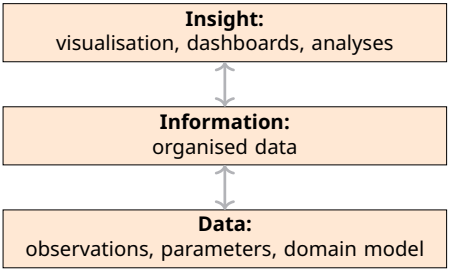

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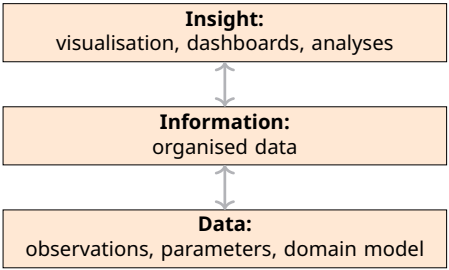

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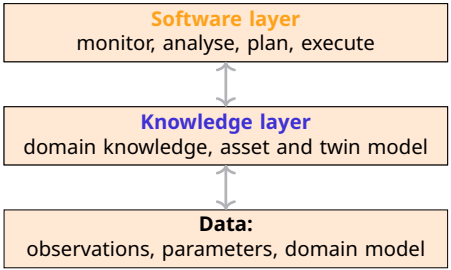

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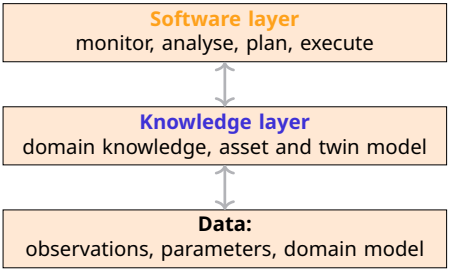

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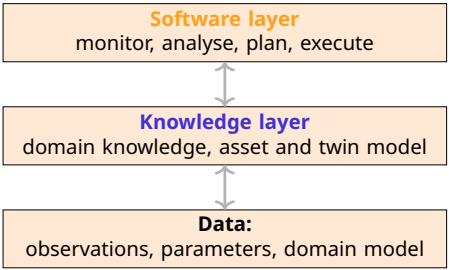

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- States of the DT cycles over time and they need to be captured

Twinning in SMOL

Programming support for twins with a behavioural and a structural layer

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Twinning in SMOL

Programming support for twins with a behavioural and a structural layer

SMOL: Semantic Model Object Language

- Smalll OO programming system
- Ontology reasoners allow querying the KB
- Used to create the digital model

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Twinning in SMOL

Programming support for twins with a behavioural and a structural layer

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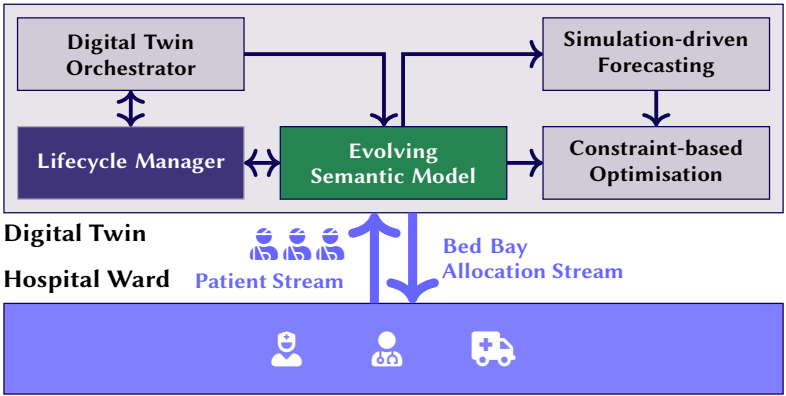
behavioural twins in SMOL

- SMOL can encapsulate (simulation) models based on the FMI standard
- Can automatically adapt the object states in the KB at runtime through semantic lifting

Contents

Architecture



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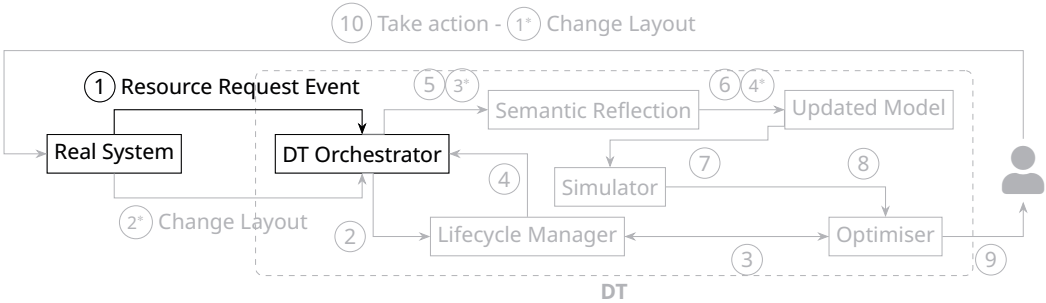

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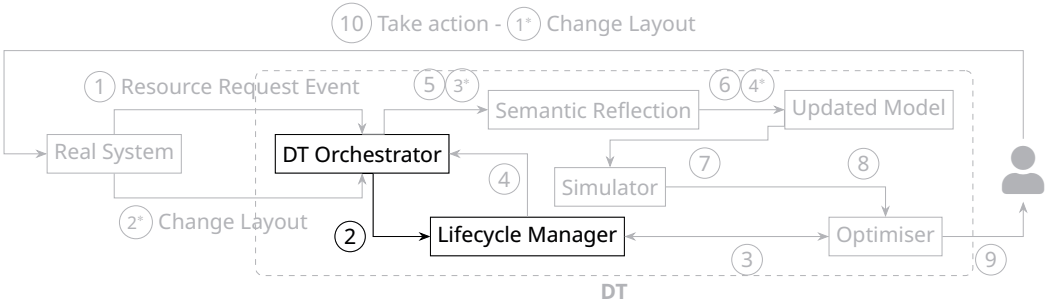

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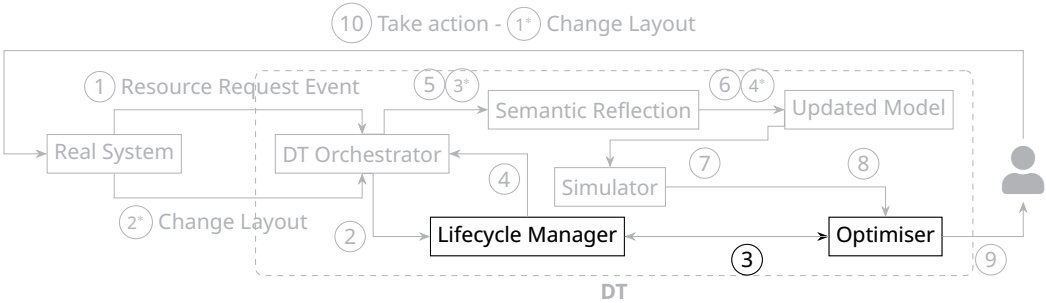

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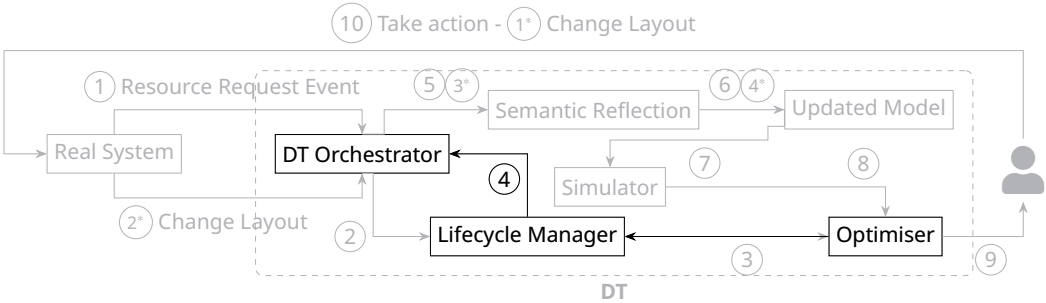

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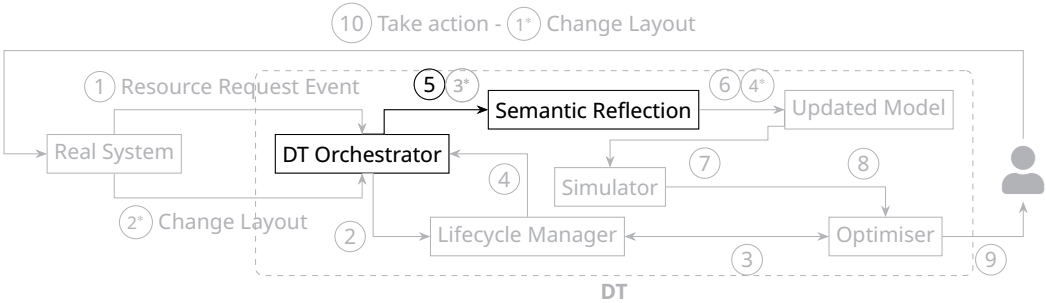

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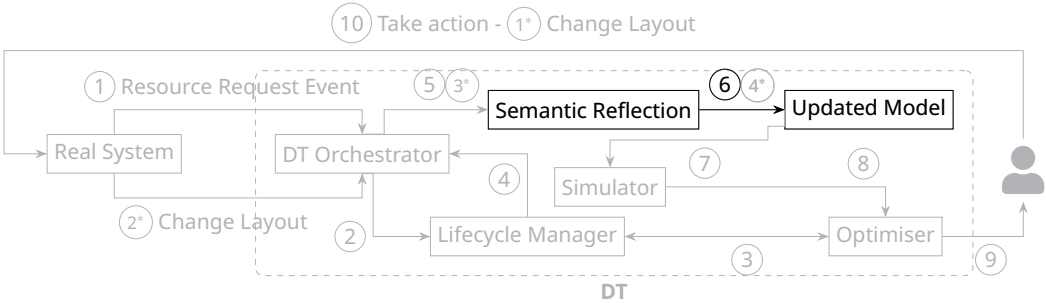

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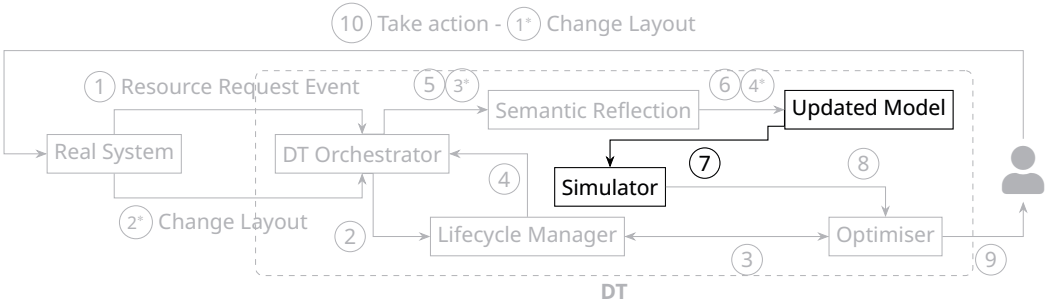

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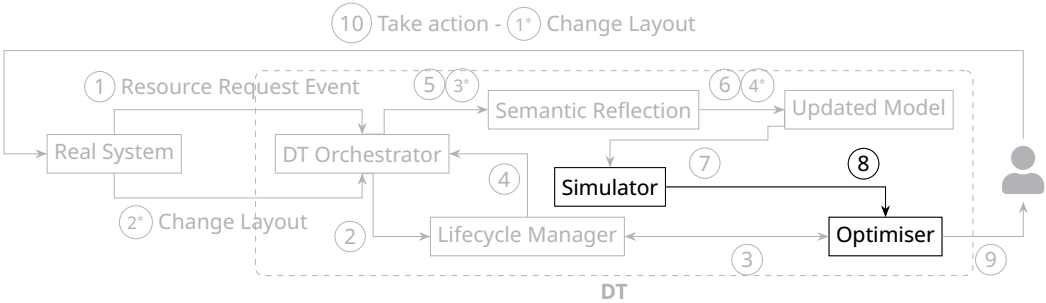

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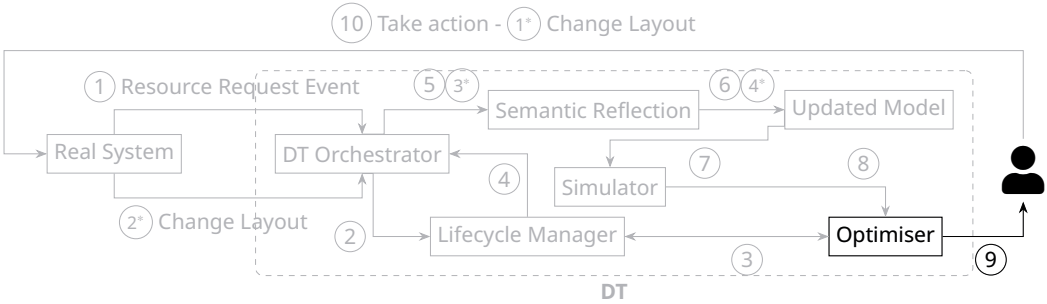

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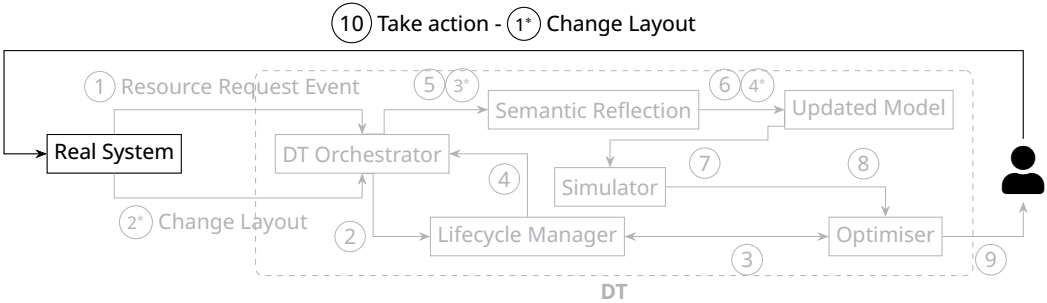

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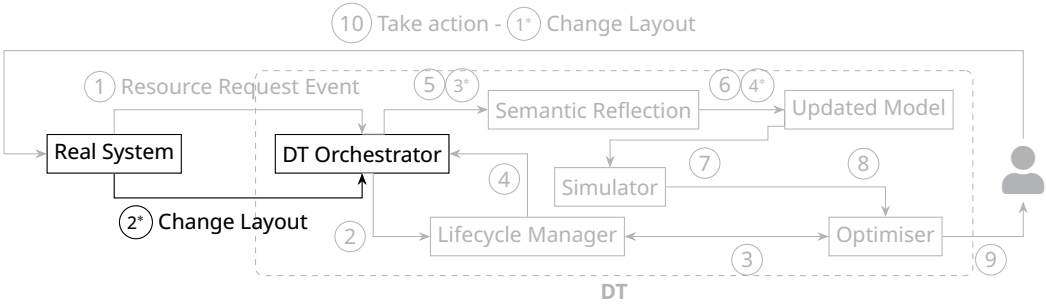

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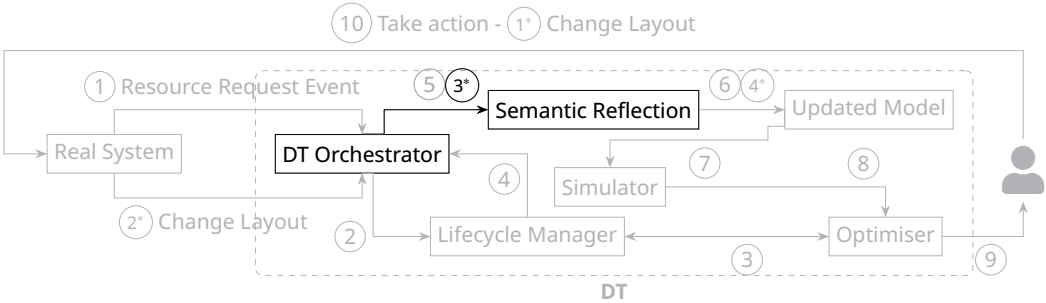

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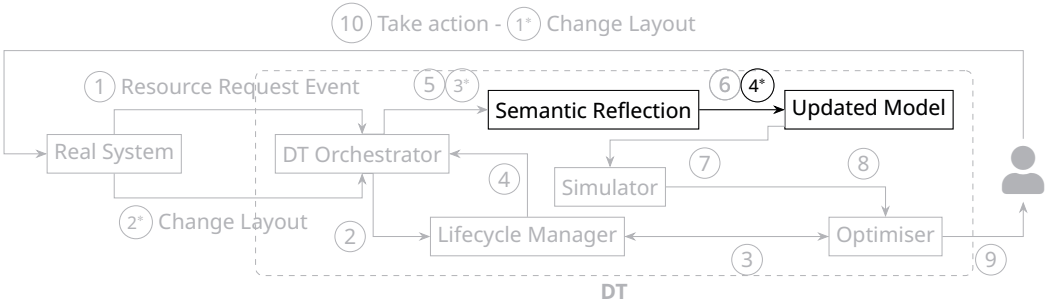

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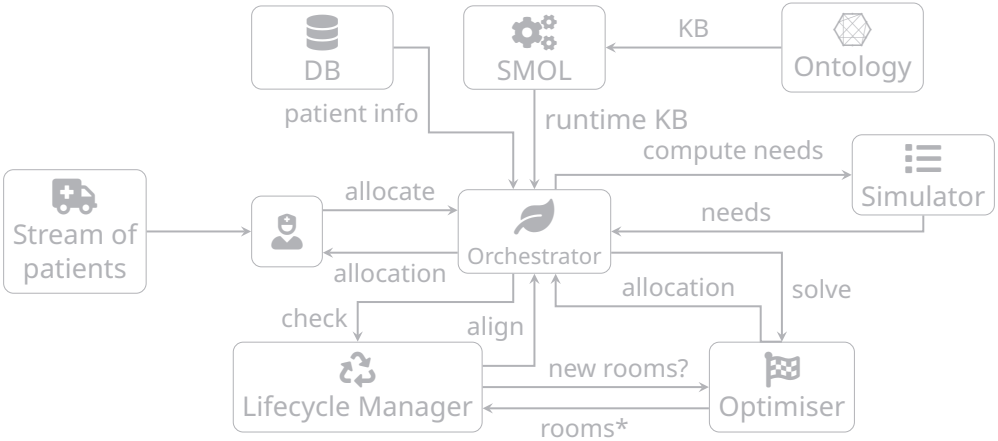
- Semantic reflection re-aligns the runtime model after structural changes
- This enables supervised autonomy: controlled self-adaptation instead of static decision support

Allocation of patients



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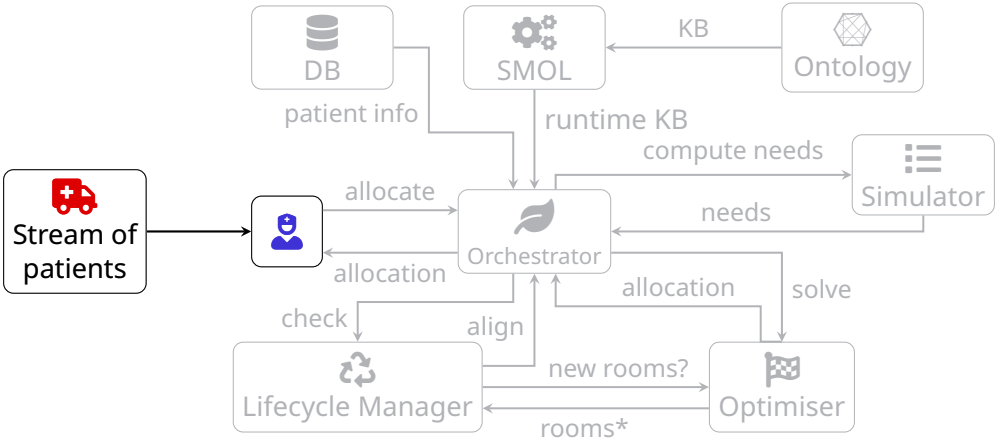

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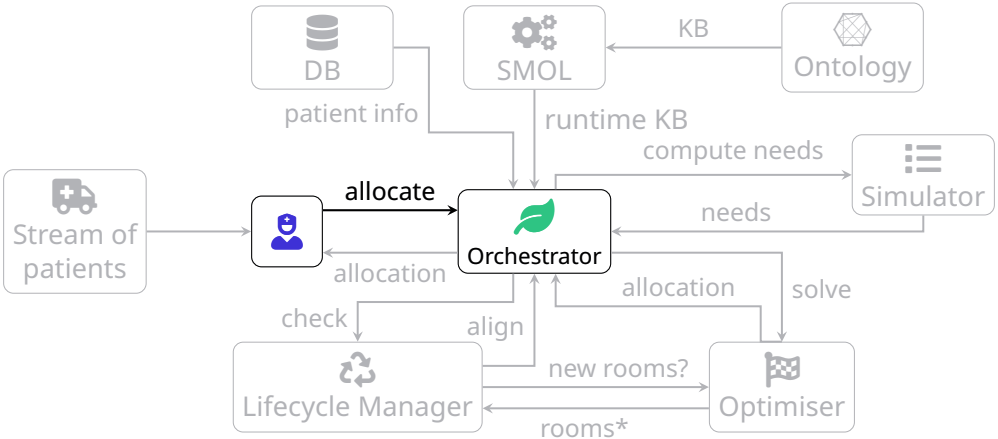

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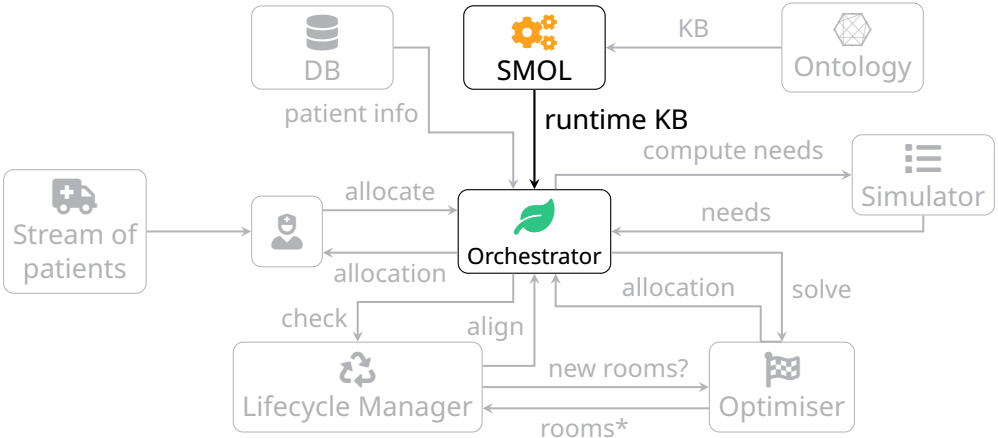

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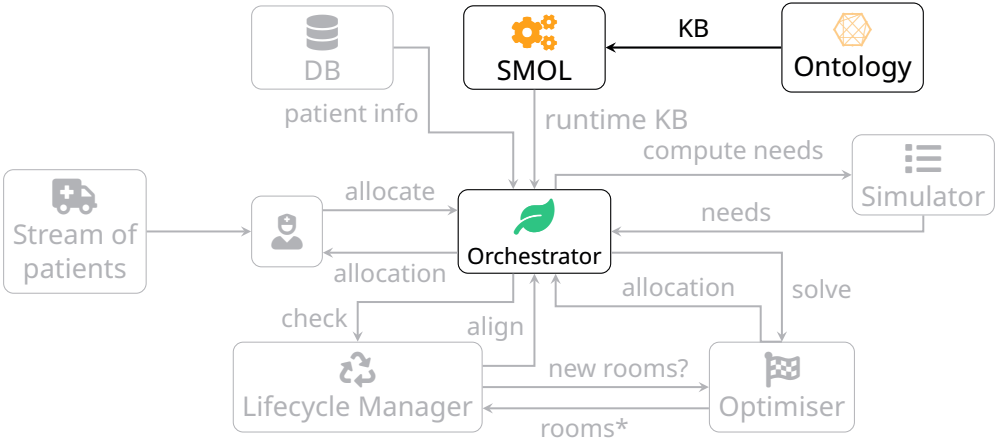

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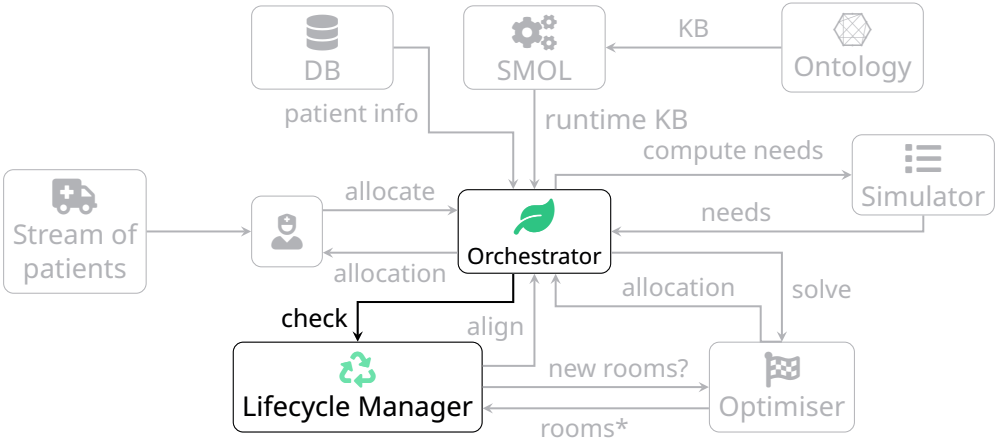

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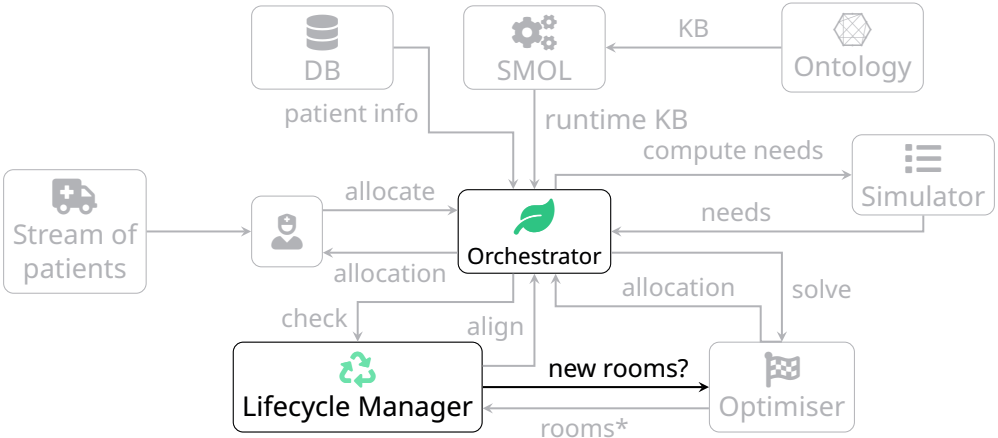

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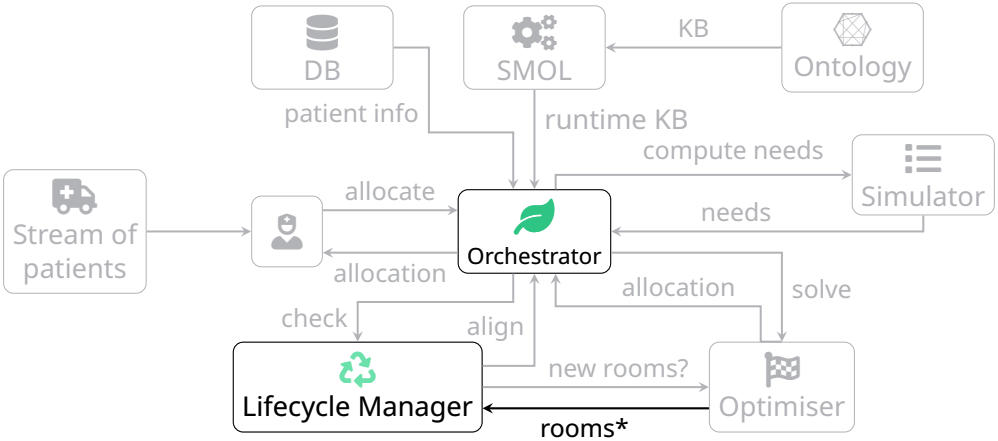

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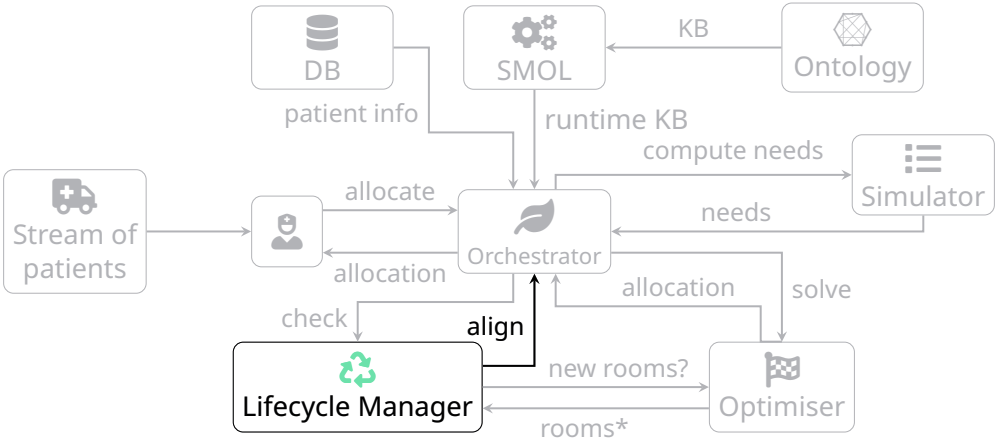

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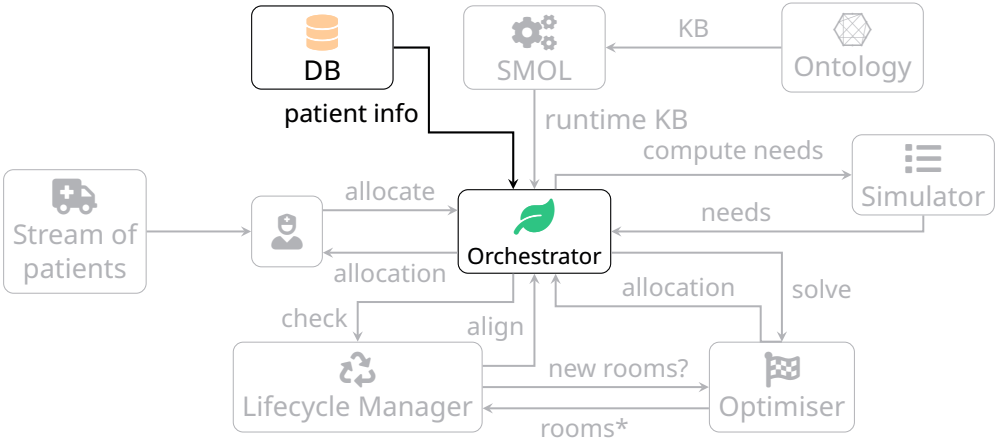

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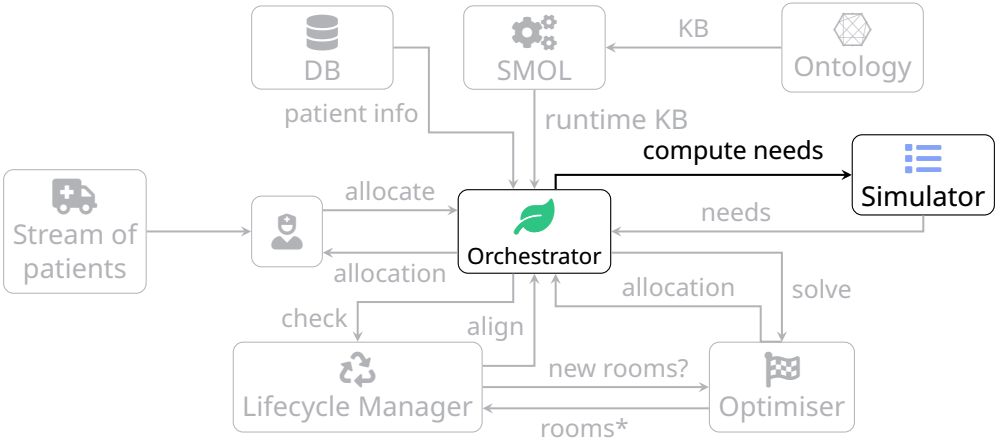

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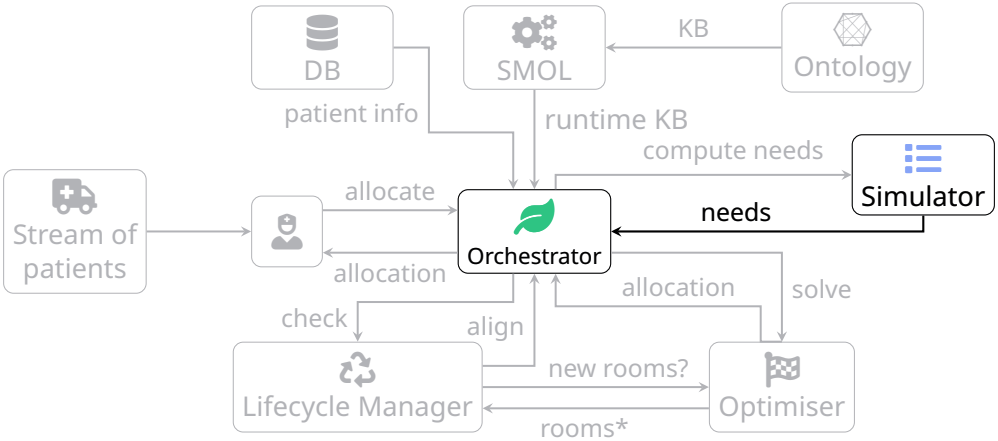

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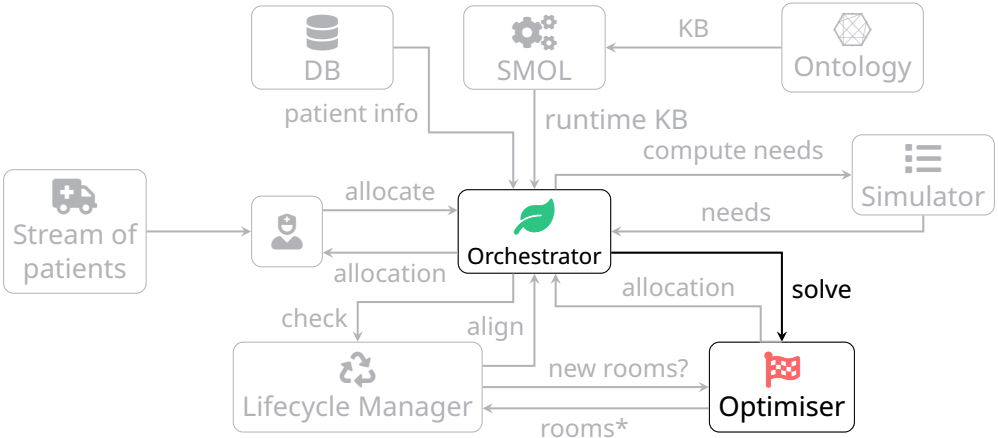

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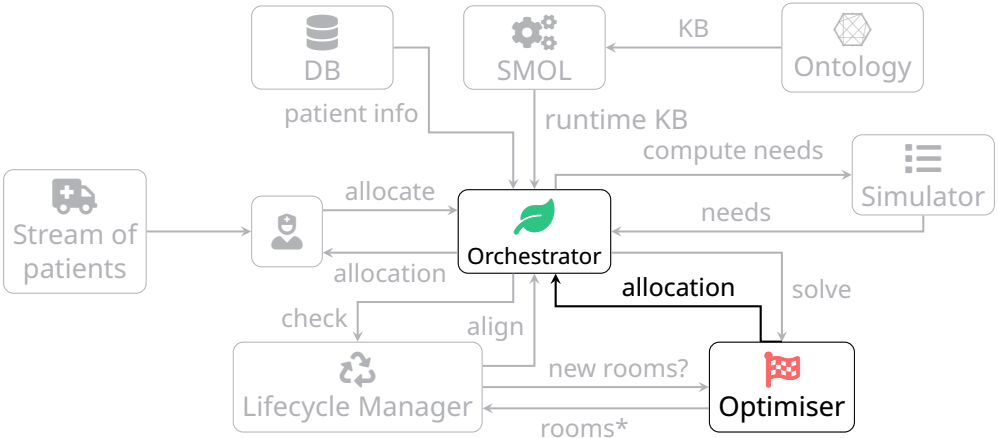

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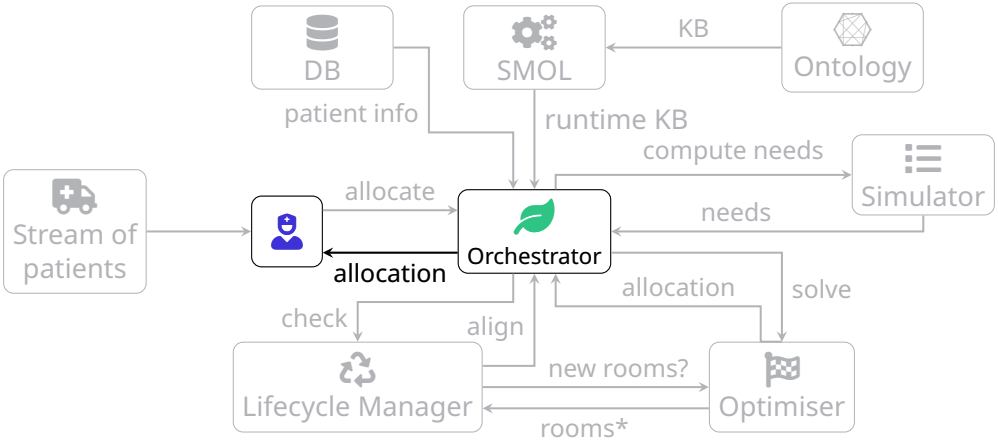
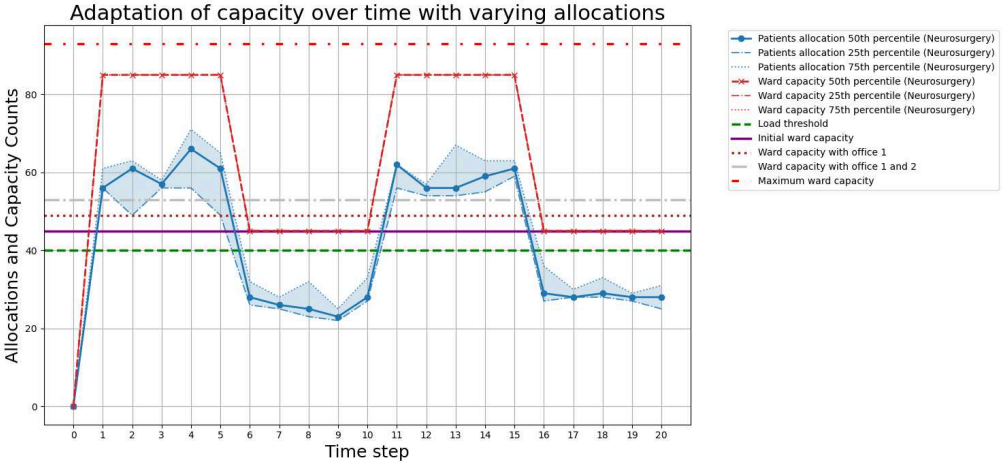


Allocation of patients (cont.)



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- Current limits: manually tuned penalties and threshold

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- Currently, SMOL provides overhead in the execution, leading to a trade-off between correctness and performance. We plan on investigate and improve the overall performance